OPEN AGENT HACKATHON 2026

FastTrack Termin

A faster route to a specialist appointment.
With evidence for every reason.

Track 02: “The Agent That Can Explain Why”
NVIDIA Nemotron prize submission

Berlin · Statutory-insurance patients · Prototype
Seed practices and waiting times are SAMPLE data. Not medical advice.

Andreas Krüger · Public repository: github.com/AndiKrue/fasttrack-termin · AGPL-3.0

PROBLEM AND TARGET USERS

Waiting is only part of the problem

Berlin statutory-insurance patients facing weeks of waiting may not know which route to try next.

Appointment service

What does the service offer, and which referral conditions apply?

Open consultation

Could an appointment-free consultation be an option?

GP referral

Could the GP help arrange the specialist appointment?

THE SOLUTION

Ask once. Compare supported routes.

Patient input

A question about location, specialty and referral situation.

The agent retrieves relevant rules, practice information and wait observations.

Patient output

Ranked routes with a reason and evidence IDs.

Each retained route exposes supporting rows with SQL or a web source URL.

The goal: identify the fastest supported route in the available data.

This prototype does not book appointments. SAMPLE waits are not live availability or a promise of access.

AGENT WORKFLOW AND ARCHITECTURE

The explanation has a data trail

1 · Plan →

Find the factual questions.

Nemotron Lightning 30B

2 · SQL →

Query three Zetaris sources.

Rules · practices · waits

3 · Recall ↓

Retrieve earlier findings.

Meterless H-MEM

6 · Remember

Store only retained routes.

Run-linked memory

5 · Verify ←

Drop unsupported routes.

Nemotron Lightning 30B

4 · Answer ←

Rank routes; cite evidence IDs.

Nemotron Ultra 550B

Between SQL and Recall: optional Firecrawl search + Lightning extraction. SQL/web rows share evidence IDs.

WHAT IS UNIQUE

A reason points to an inspectable source

Claim

A route and its factual explanation.

No invented appointment promise.

Evidence ID

`E1`

Illustrative ID

Evidence object

`SQL + row` or `URL + row`

Inspect the exact query result or a web fact and its source URL.

The verifier is instructed to keep a route only when every factual statement is supported by its cited rows.

Unknown IDs are removed; routes need a valid citation and a keep verdict. This is an LLM check, not a correctness guarantee.

Three sources. One SQL surface.

Source	What the agent retrieves
`FT_RULES.rules`	Route rules, applicability and legal references
`FT_PRACTICES.practices`	Practice directory and consultation attributes
`FT_WAITS.waits`	Channel-specific waiting-time observations

Zetaris Lightning: Docker Compose platform; queries over JDBC. `zetaris/register.py` registers the seed CSV tables.

Local fallback: SQLite with the same schemas. Seed practices/waits are SAMPLE. Optional Firecrawl adds real web practice facts with URLs; the answer prefers these rows.

METERLESS H-MEM

Memory keeps a link to its origin

Recall before answering

A TypeScript bridge queries H-MEM for earlier findings relevant to the question.

Recalled findings provide context; the answer still needs current evidence rows.

Remember after verification

Retained route and reason become factual memory with a run tag and `fasttrack:<run_id>` source.

H-MEM persists state and a ledger.

Current provenance boundary: H-MEM stores text and the run link. Full SQL/rows are in the returned evidence; the JSONL fallback stores the full record.

Four roles, with separate model choices

Role	Configured model	Responsibility
Planner	Lightning 30B	Select factual questions
SQL	Lightning 30B	SELECTs and web search query
Verifier	Lightning 30B	Route checks; web fact extraction
Answer	Ultra 550B	Rank routes and explain why

Python: agent, HTTP app, `httpx`, JDBC via `jaydebeapi`.
Node / TypeScript: `tsx` bridge to Meterless H-MEM.
Browser: HTML/CSS/JavaScript question and evidence view.

NVIDIA API · Optional Firecrawl: cloud or self-hosted · No measured gains claimed.

PRODUCT VISUALS

Show the route, then inspect the reason

Ask and compare

SCREENSHOT PLACEHOLDER A

Question + ranked route cards

Insert an actual app capture.
Keep the SAMPLE label visible.

Inspect the evidence

SCREENSHOT PLACEHOLDER B

Evidence ID + SQL / URL + row

Insert an actual app capture.
Include source / memory backend.

Placeholders only. No screenshots or live-run results are represented on this slide.

FUTURE WORK

From sample routes to real access

Real practice data

Expand web coverage; replace SAMPLE waits with maintained data.

Track freshness and source provenance.

Booking integration

Connect supported routes to booking services.

Confirm availability and obtain patient approval before booking.

Other cities

Extend directories and validate local route information.

Keep the same evidence and verifier contract.

Next validation: live integrations, verifier failure cases, and complete row provenance in H-MEM.

Public source: github.com/AndiKrue/fasttrack-termin